IDAT7215
Computer Programming for Product Development and Applications

Lecture 1-1: Introduction to Programming

Dr. Zulfiqar Ali

What is a Programming Language?

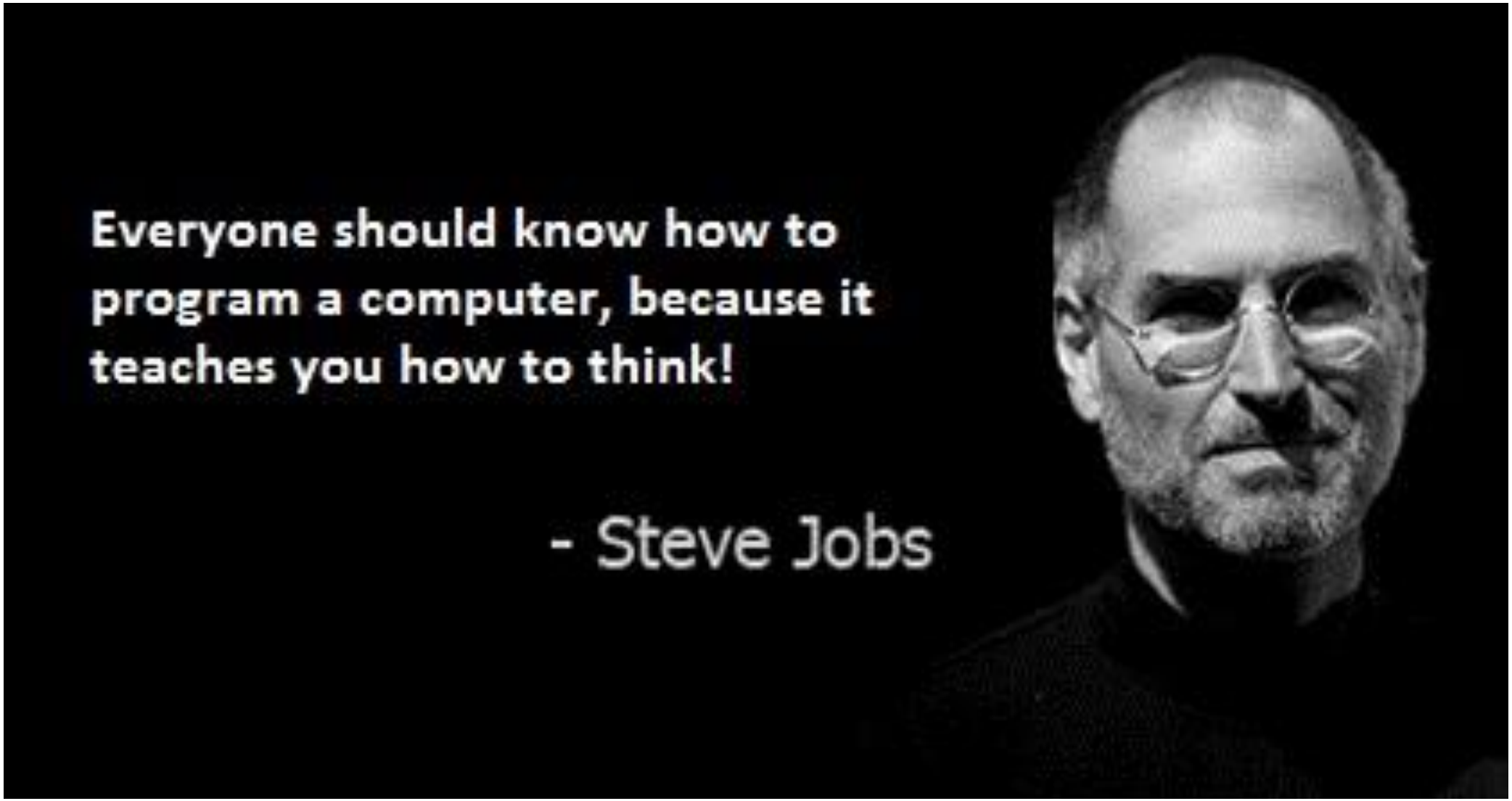


A programming language is a set of written symbols/rules that instructs the computer hardware to perform specific tasks.

Why Programming Is Important?

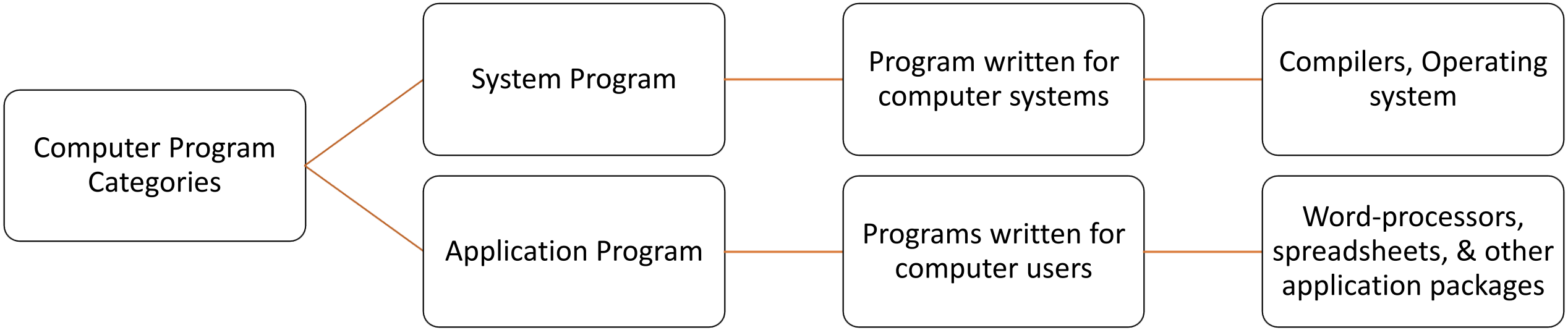
- Programming helps you understand computers.
- A computer is only a tool. If you learn how to write simple programs, you will gain more knowledge about how a computer works.
- Writing a few simple programs increases your confidence level.
- Many people find great personal satisfaction in creating a set of instructions that solve a problem.
- Learning programming lets you find out quickly whether you like programming and whether you have the analytical turn of mind programmers need.
- Even if you decide that programming is not for you, understanding the process certainly will increase your appreciation of what programmers and computers can do.

Why Programming Is Important?

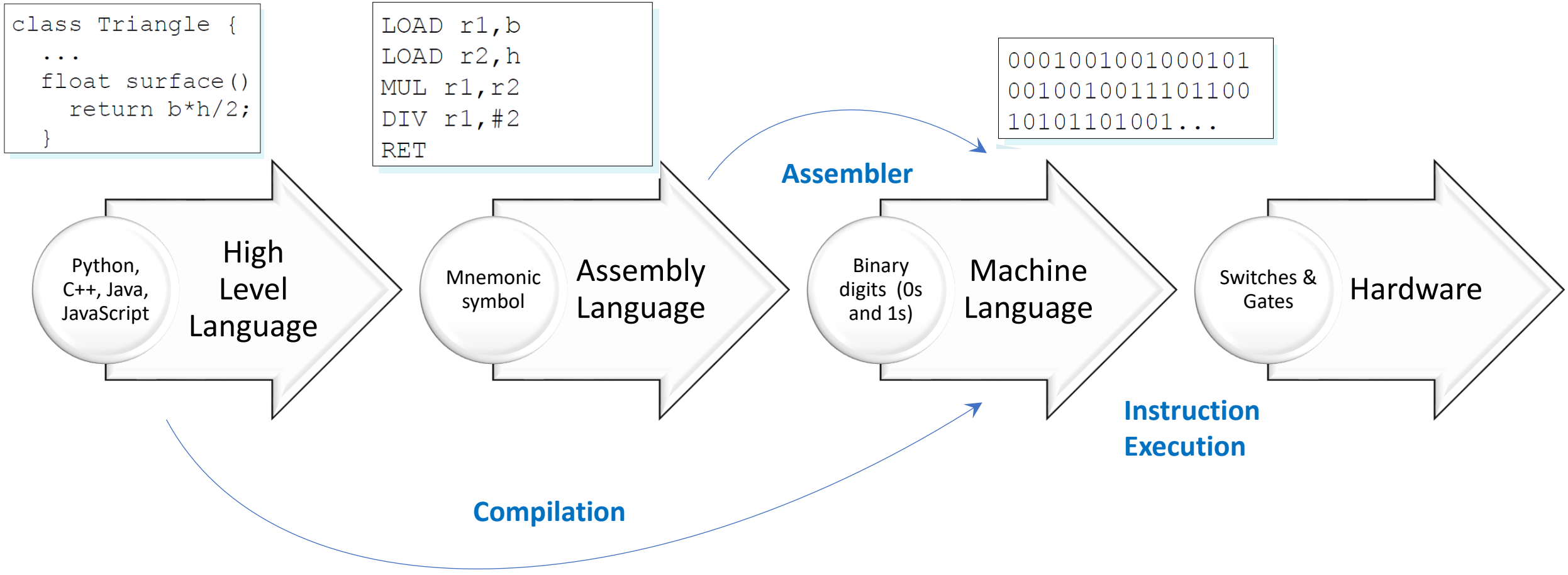


Why are there so many programming languages?

- Why does some people speak French?
- Programming languages have evolved over time as better ways have been developed to design them.
  - First programming languages were developed in the 1950s.
  - Since then, thousands of languages have been developed.
- Different programming languages are designed for different types of programs.

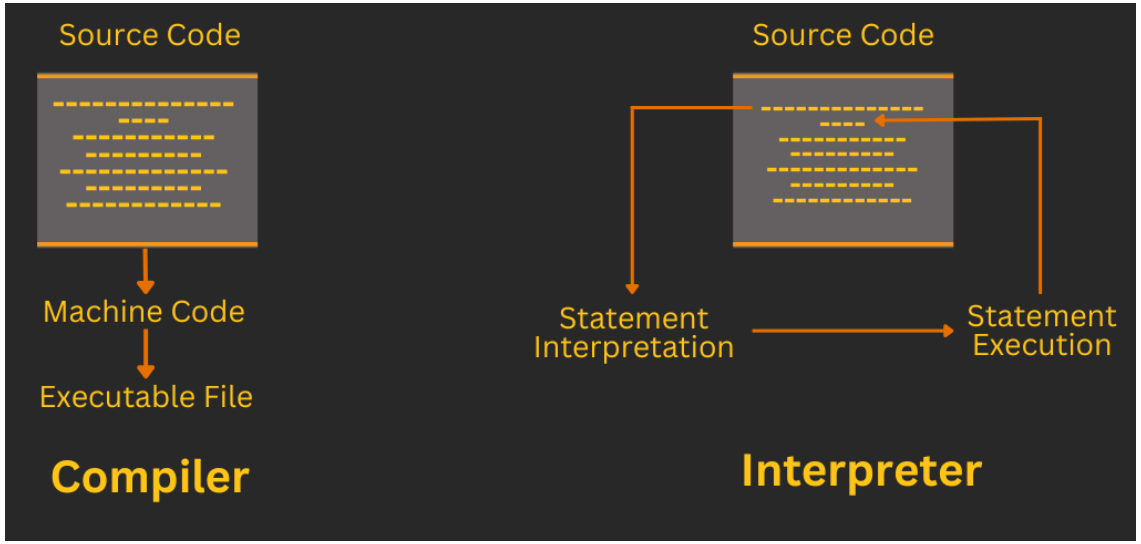

Computer Program Categories
System Program
Application Program
Program written for computer systems
Programs written for computer users
Compilers, Operating system
Word-processors, spreadsheets, & other application packages

Computer Languages



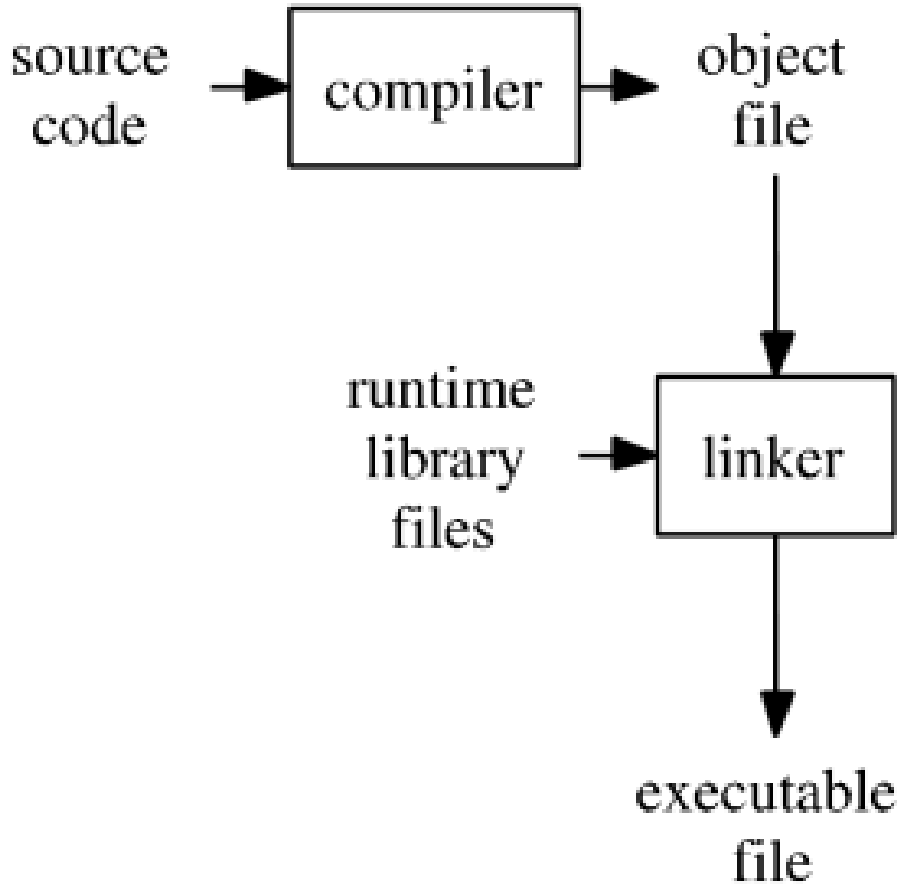
Compilers & Programs

- **Compiler**
  - Compiler translates the entire source code into machine code before execution.
  - Faster execution since no translation is needed during runtime.
  - E.g., C, C#, C++
- **Interpretator**
  - An interpreter translates code line by line during execution, making it easier to detect errors.
  - Potentially slowing down the program.
  - E.g., Python, Ruby, JavaScript.



Compilers & Programs

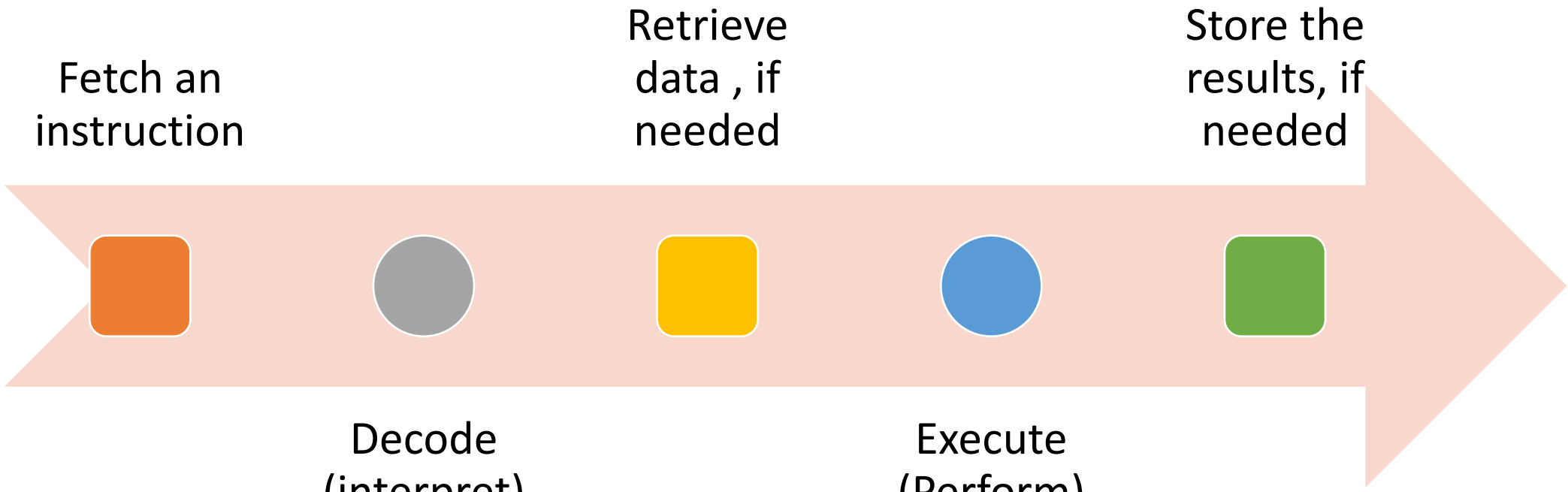
- **Source program**
  - The form in which a computer program, written in some formal programming language, is written by the programmer.
  - Can be compiled automatically into object code or machine code or executed by an interpreter.
- **Object program**
  - Output from the compiler
  - Equivalent machine language translation of the source program
  - Files usually have extension '.obj'



Compilers & Programs

- **Executable program**
  - Output from linker/loader.
  - Machine language program linked with necessary libraries & other files.
  - Files usually have extension '.exe'
- **What is Linker?**
  - A program that pulls other programs together so that they can run.
  - Most programs are very large and consist of several modules.
  - Even small programs use existing code provided by the programming environment called libraries.
  - The linker pulls everything together, makes sure that references to other parts of the program (code) are resolved.

Program Execution



Steps taken by the CPU to run a program
(instructions are in machine language)

Program Errors

Syntax Error

- Error in grammar of the language

```
1  number = 55
2  if number == 55
3      print('True')
4  else
5      print('False')
6
```

Missing Colon

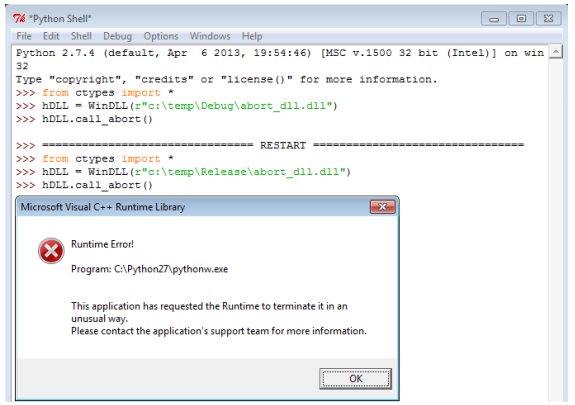
```
Shell
>>> %Run -c $EDITOR_CONTENT
  Traceback (most recent call last):
    File "<string>", line 2
      if number == 55
                   ^
  SyntaxError: expected ':'
>>>
```

Logical Error

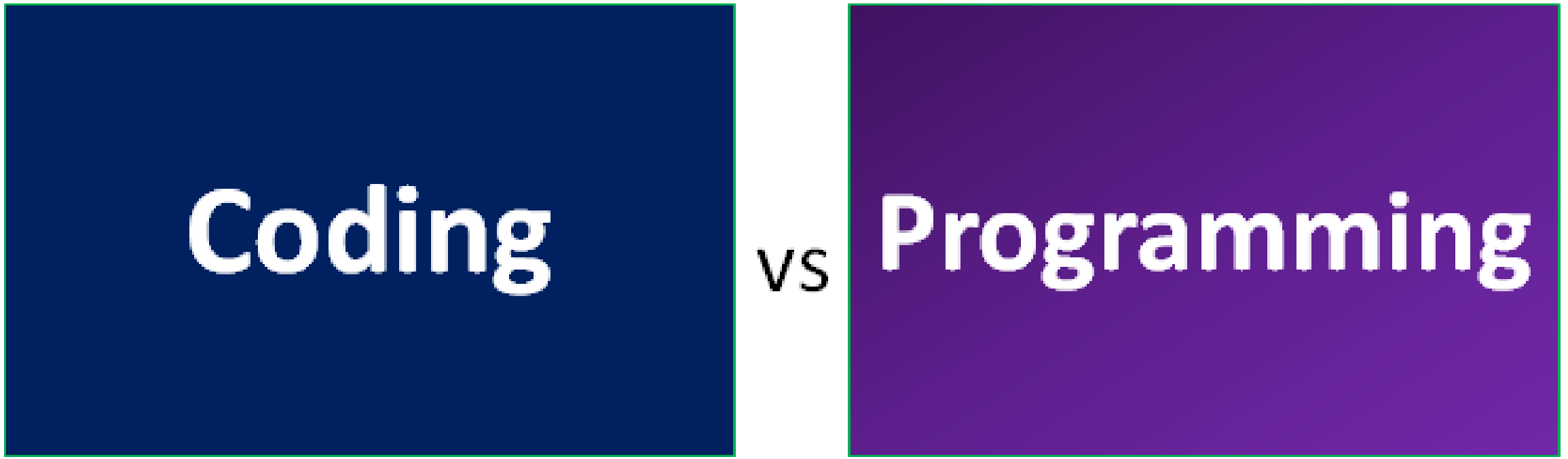
```
x = 2
y = 14
average = x + y / 2
print(average)
```

- Deliver incorrect results
- Incorrect usage of parentheses

Runtime Error



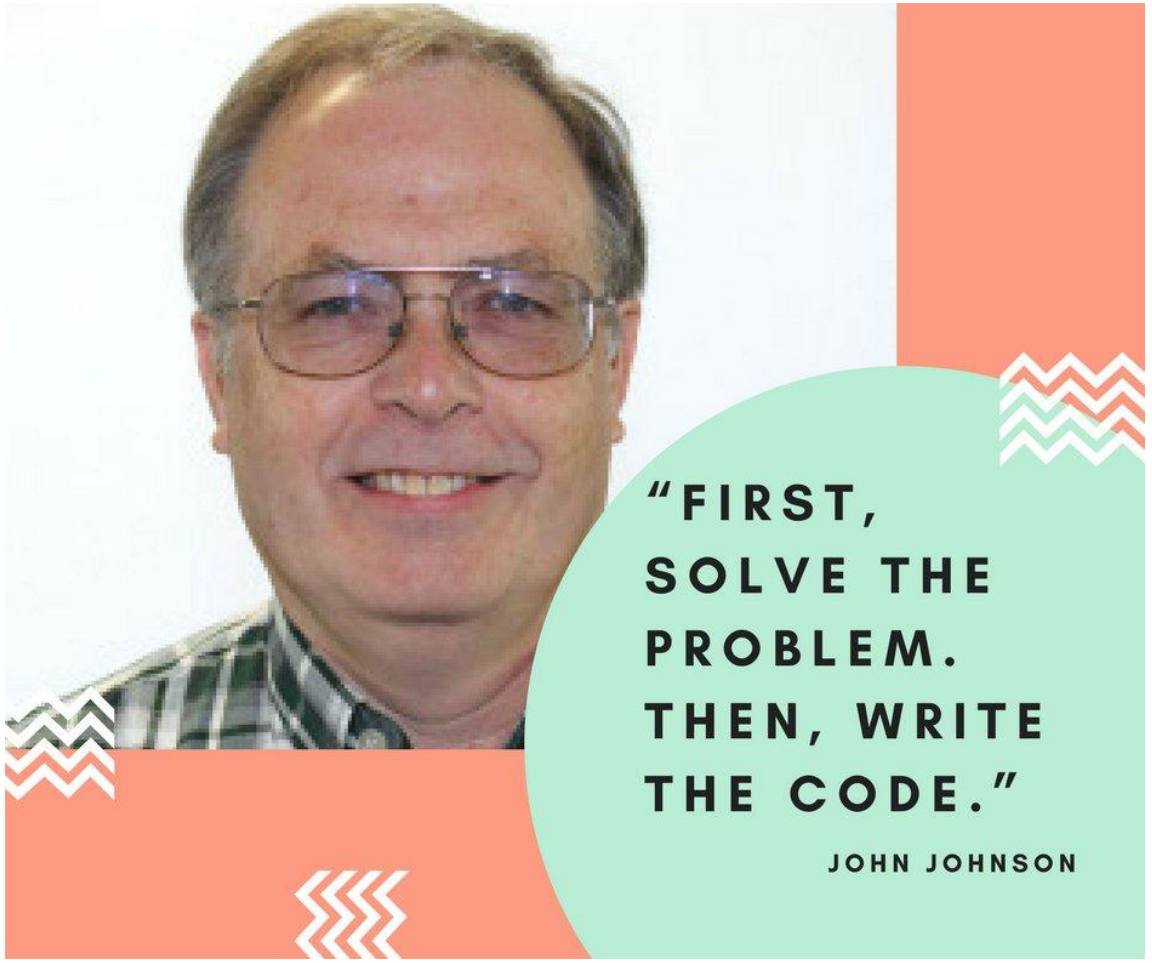
- Divide by zero
- Invalid input data

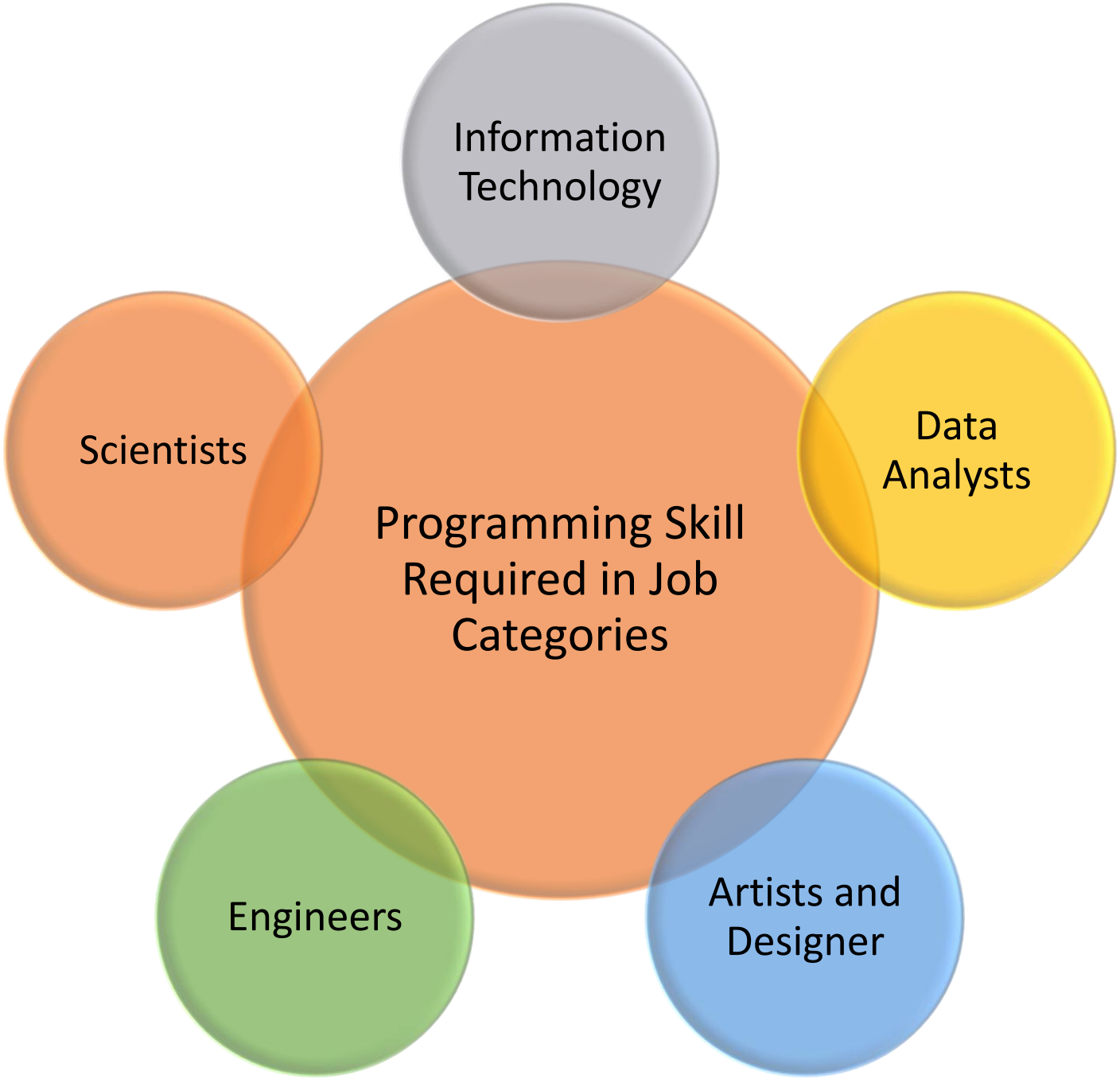

Coding
vs
Programming

Programming is a process

- Problem Identification and Solving Phase
- Maintenance Phase
- Implementation Phase

All three of them are important for computer programmers or software developers to be able to solve problems. A lot of developers misunderstood that the programming process is only algorithm implementation, which is incorrect.




Information Technology
Scientists
Programming Skill Required in Job Categories
Data Analysts
Engineers
Artists and Designer

What's most in demand?

The report found that the highest demand was for programming languages with broad applicability. Other skills in demand include:





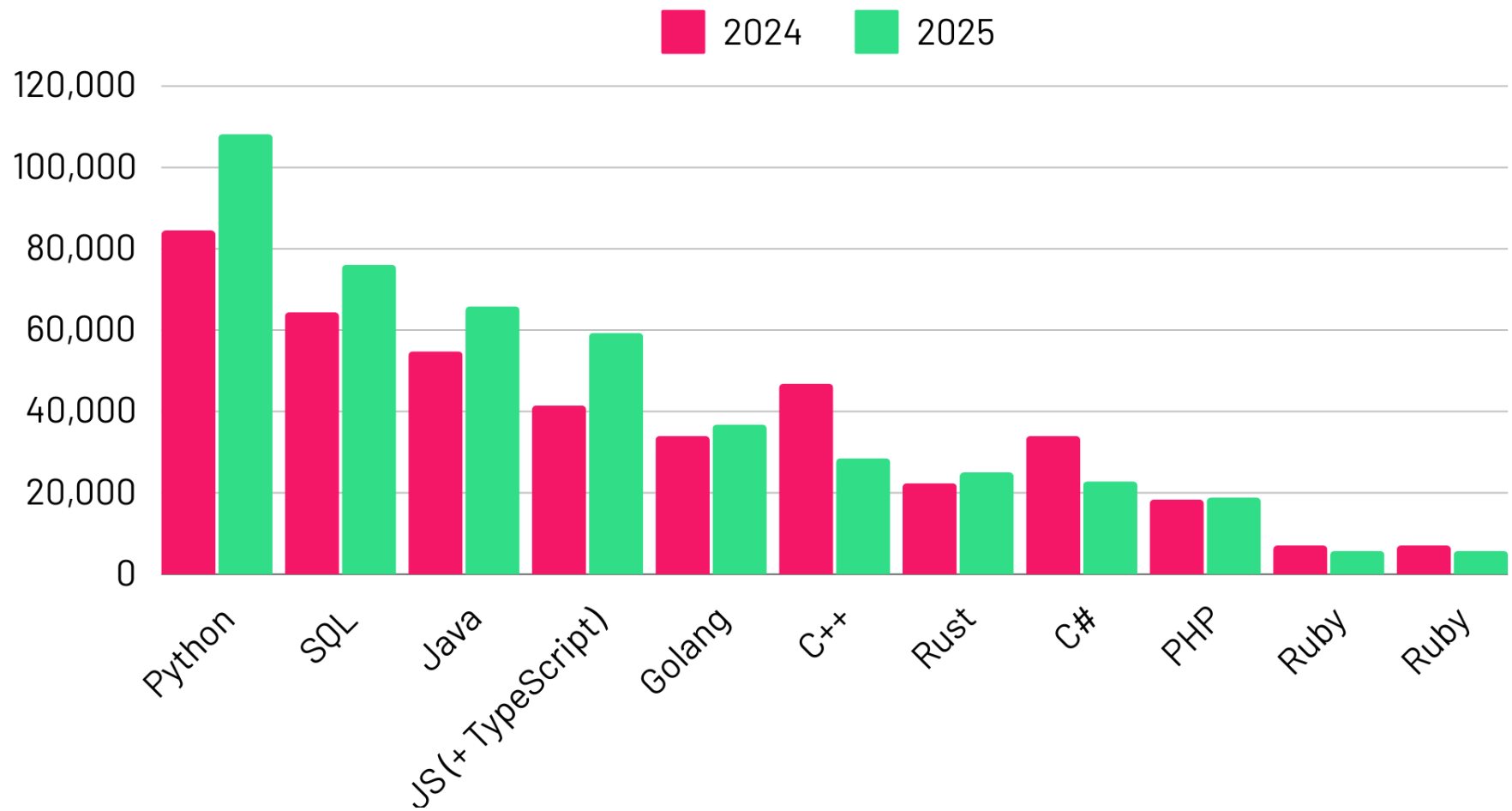






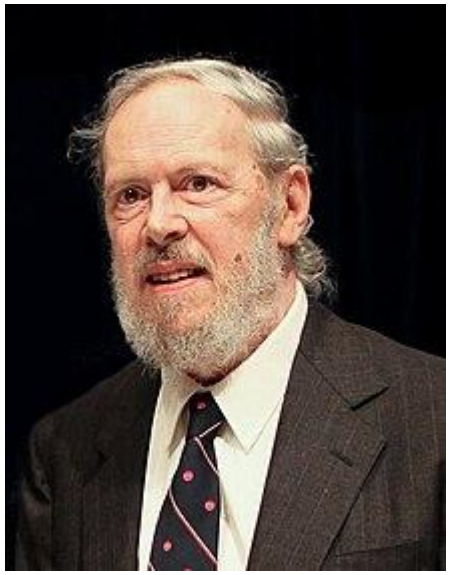
2025 PROGRAMMING LANGUAGE BREAKDOWN

of open jobs available for each language in US in 2025 (Compared to 2024)



https://zerotomastery.io/blog/best-programming-languages-to-learn/

C Language

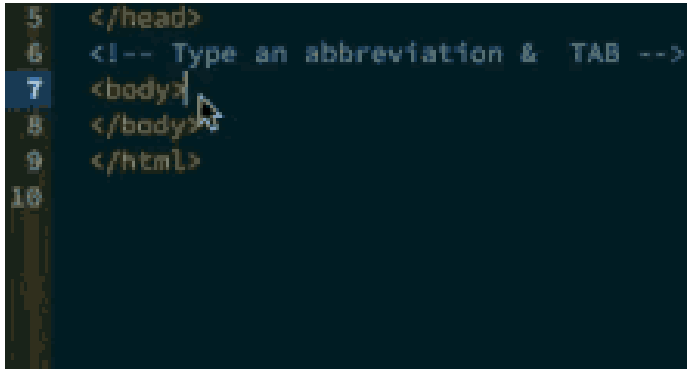




- C is a general-purpose, high-level language that Dennis M. Ritchie originally developed to develop the UNIX operating system at Bell Labs.
- C was originally first implemented on the DEC PDP-11 computer in 1972.
- In 1978, Brian Kernighan and Dennis Ritchie produced the first publicly available description of C, now known as the K&R standard.
- The UNIX operating system, the C compiler, and all UNIX application programs have been written in C.

Structure of C Program

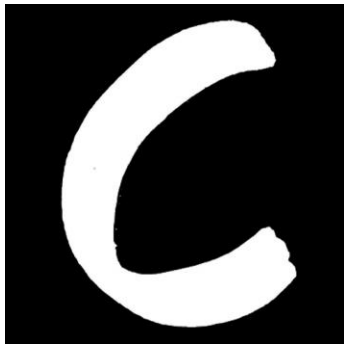
	1	`#include <stdio.h>`	Header
	2	`int main(void)`	Main
BODY	3	`{`	
BODY	4	`printf("Hello World");`	Statement
BODY	5	`return 0;`	Return
BODY	6	`}`	



C Language

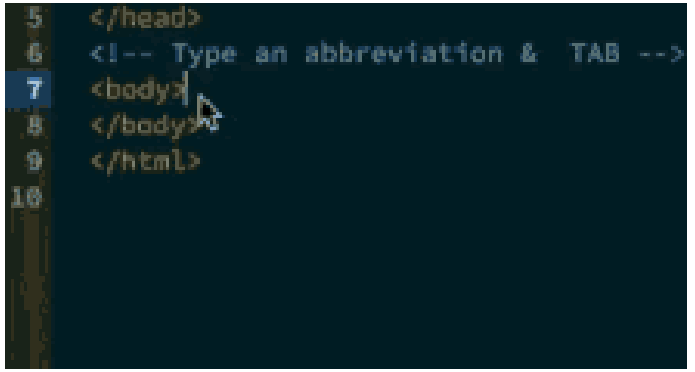
- C has now become a widely used professional language for various reasons.
  - Easy to learn
  - Structure language
  - It produces efficient programs
  - It can handle low-level activities.
  - It can be compiled on a variety of computer platforms.



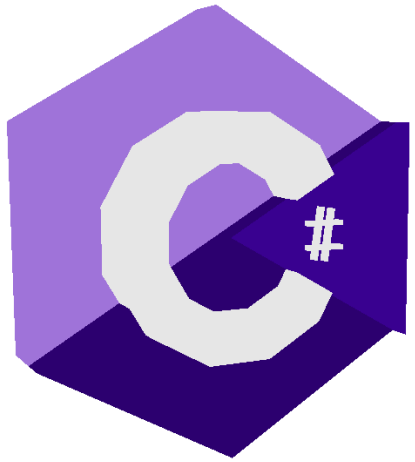


Structure of C Program

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C# Language



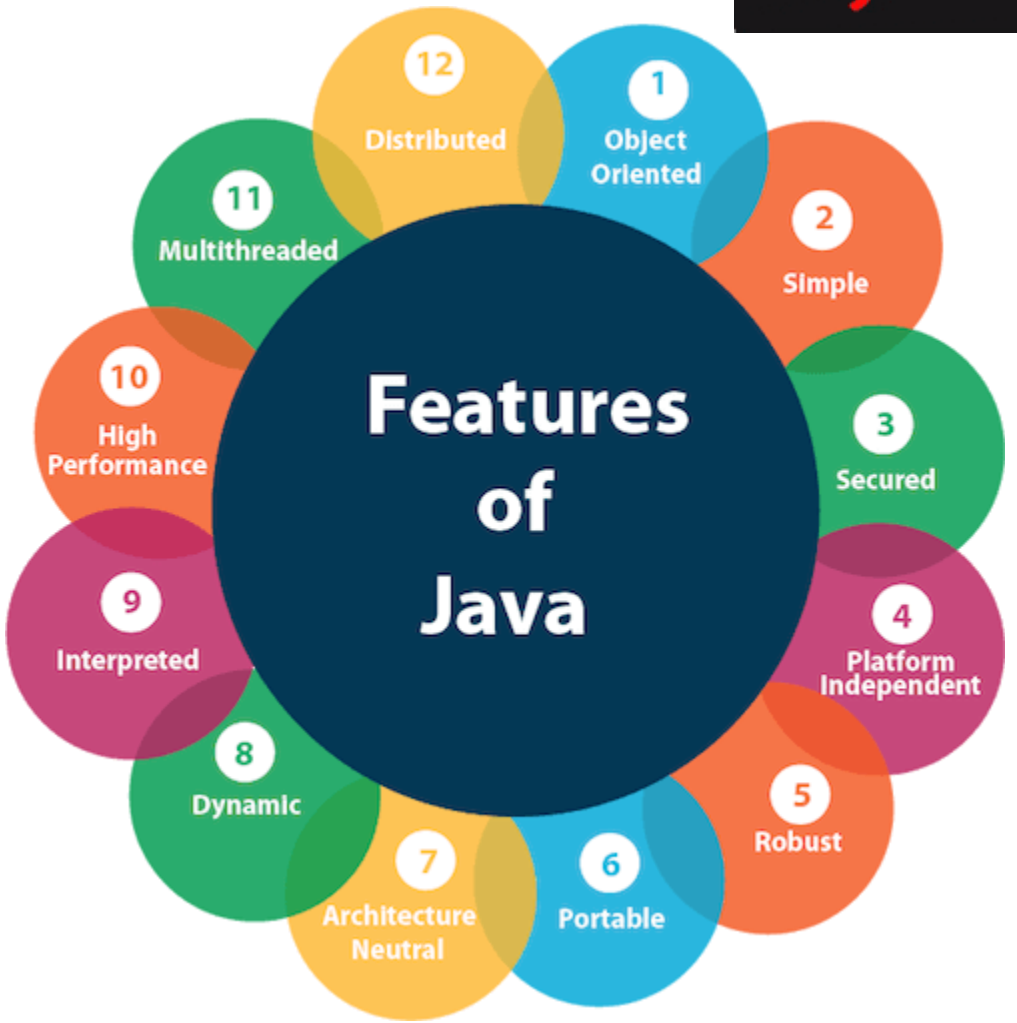
- Nowadays, C# is the most popular language in the world.
- Microsoft developed C# within its .NET framework initiative and later approved as a standard by ECMA.
- C# programming language is a general-purpose OOPS-based programming language.
- The C# development team was led by “Anders Hejlsberg” in 2002.
- C# programming language is one of the languages designed for Common Language Infrastructure (CLI).



Java Language

- Java is a popular general-purpose programming language and computing platform. It is fast, reliable, and secure. According to Oracle, the company that owns Java, Java runs on 3 billion devices worldwide.

- Considering the number of Java developers, devices running Java, and companies adapting it, it is safe to say that Java will be around for many years to come.



JavaScript Language

- JavaScript was initially created to “make web pages alive”.
- The programs in this language are called scripts. They can be written right in a web page’s HTML and executed automatically as the page loads.
- Scripts are provided and executed as plain text. They do not need special preparation or compilation to run.
- In this aspect, JavaScript is very different from another language called Java.





Python Language



- Python is a widely used general-purpose, high-level programming language.

- It was initially designed by Guido van Rossum in 1991 and developed by Python Software Foundation. It was mainly developed for emphasis on code reliability, and its syntax allows programmers to express concepts in fewer lines of code.

- Python is a programming language that lets you work quickly and integrate systems more efficiently.



Real World Examples



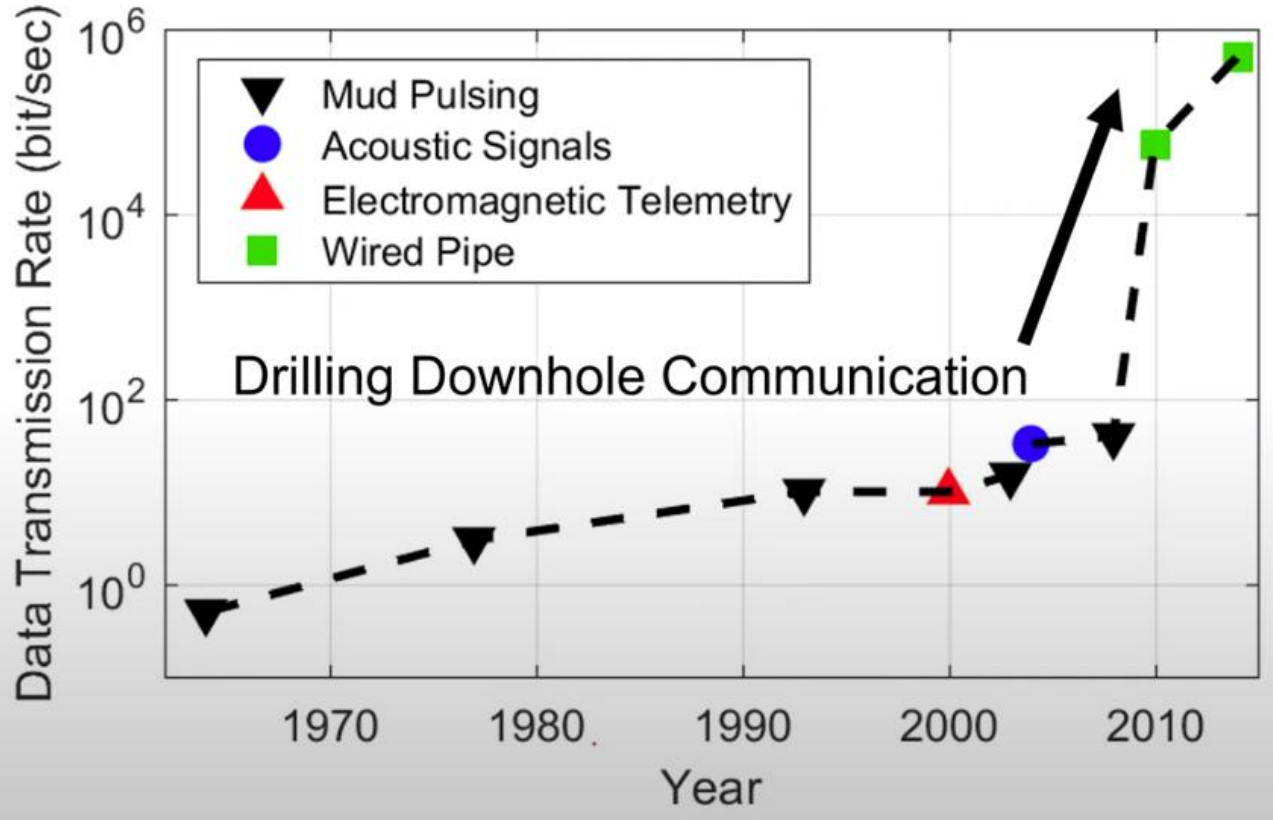
25 GB/hour



150,000 points/sec



51,200 GB/hr



Hedengren, J. D., Eaton, A. N., Overview of Estimation Methods for Industrial Dynamic Systems, Springer, 2017, DOI: 10.1007/s11081-015-9295-9.





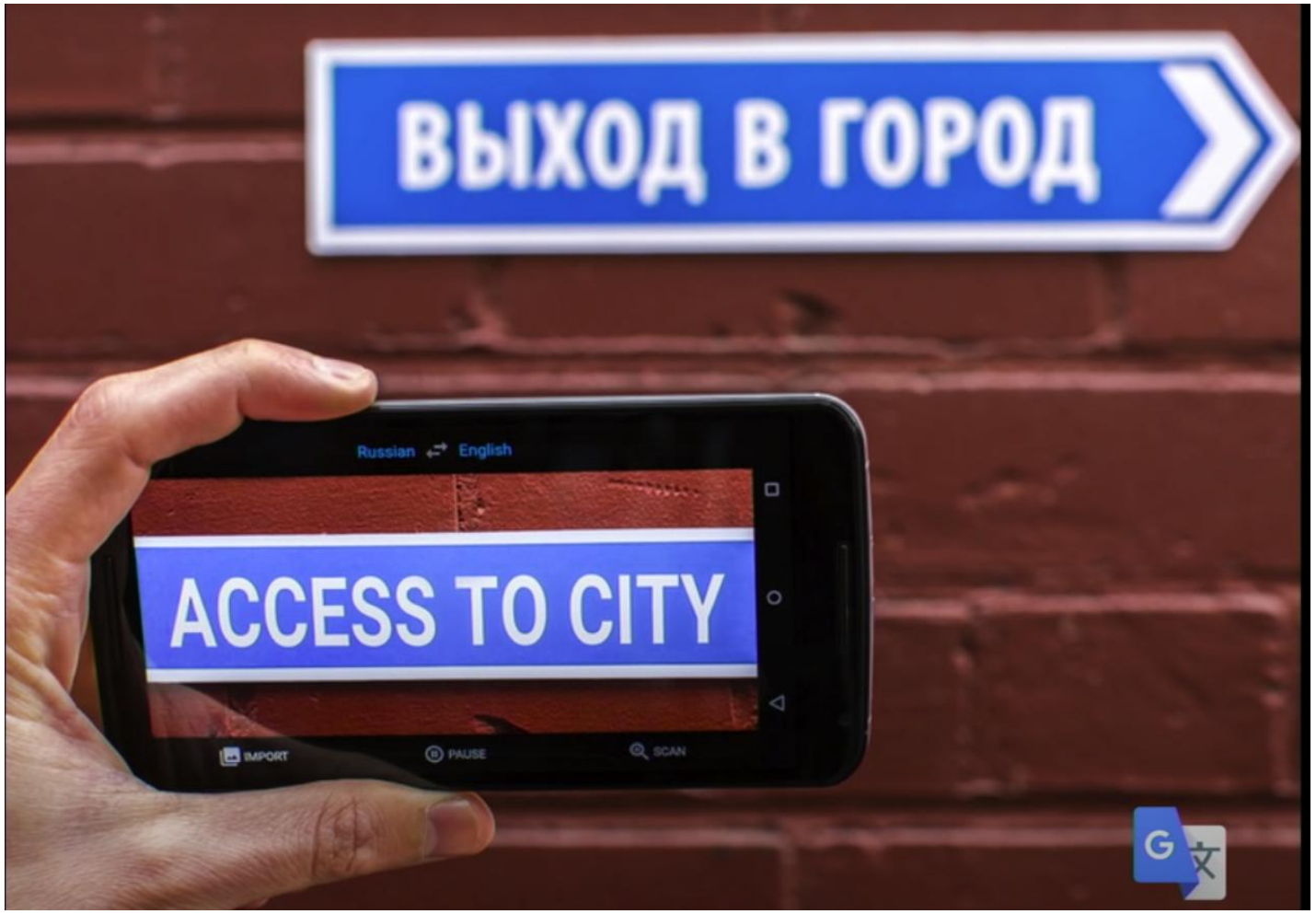


Sensors, Actuators, and Algorithms





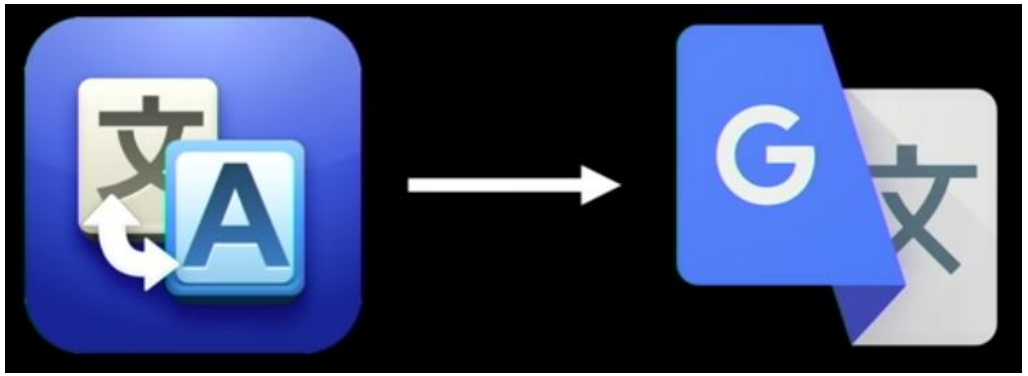






500 GB

30 million personal computer



Google Data 10-15 Exabyte

1 EB = 1,000, 000 TB
1 TB = 1000 GB

https://www.youtube.com/watch?v=SN2BZswEWUA&ab_channel=TEDxTalks

Jobs in Data Analytics

Top 10 Best Jobs in America in 2020

Rank	Job Title	Median Base Salary	Job Satisfaction	Job Openings
1	Front End Engineer	$105,240	3.9	13,122
2	Java Developer	$83,589	3.9	16,136
3	Data Scientist	$107,801	4.0	6,542
4	Product Manager	$117,713	3.8	12,173
5	Devops Engineer	$107,310	3.9	6,603
6	Data Engineer	$102,472	3.9	6,941
7	Software Engineer	$105,563	3.6	50,438
8	Speech Language Pathologist	$71,867	3.8	29,167
9	Strategy Manager	$133,067	4.3	3,515
10	Business Development Manager	$78,480	4.0	6,560

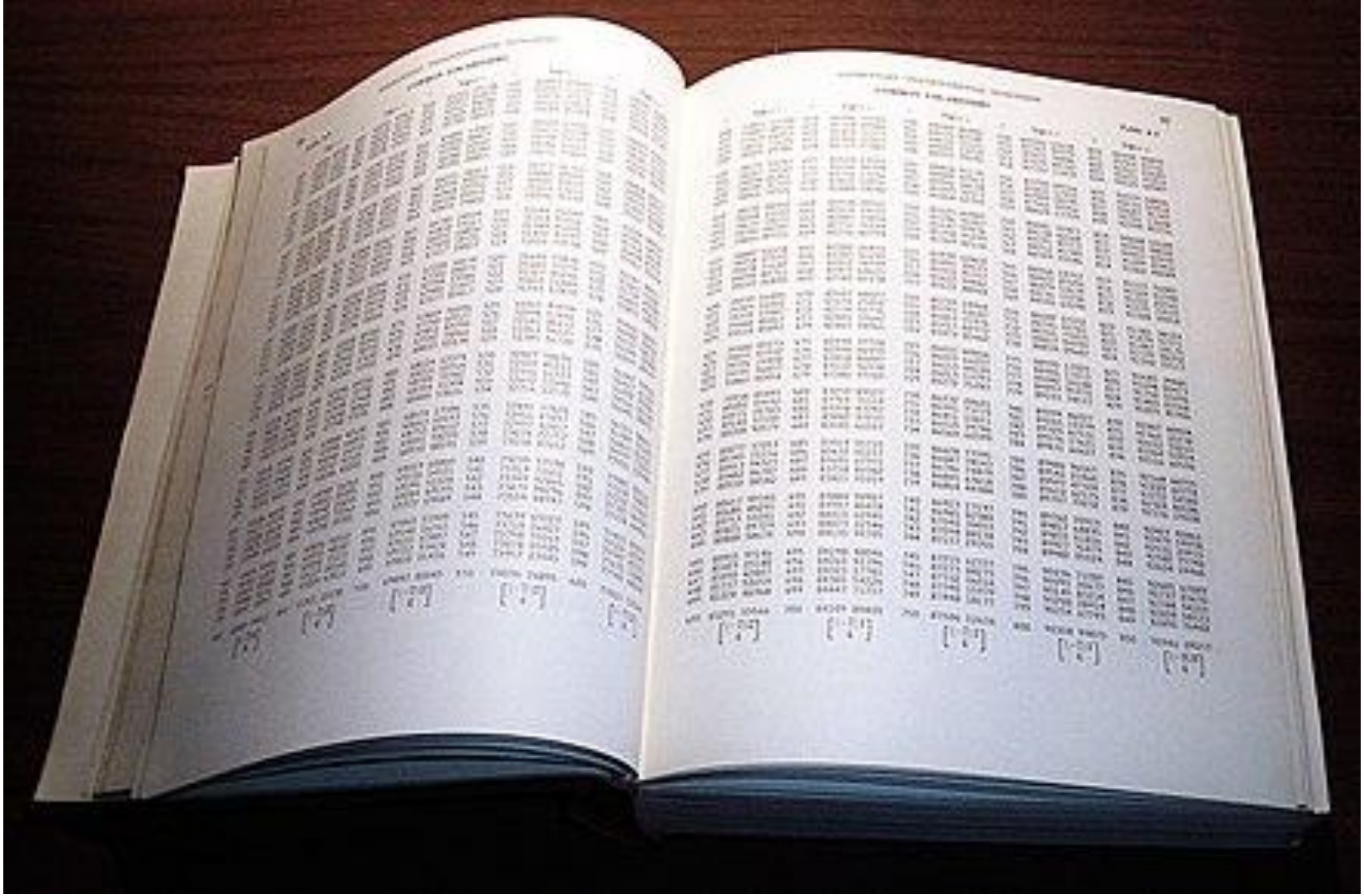
Indeed's best jobs of 2019

Rank	Job title	Average Y/Y % change in the share of job postings, 2015–2018	Average salary	Number of postings per 1M total jobs, 2018
1	Machine learning engineer	629%	$102,500	173
2	Veterinarian	245%	$87,000	154
3	Full-stack developer	152%	$87,500	1,764
4	Data scientist	128%	$98,190	569
5	Dentist	119%	$124,677	139
6	Senior back-end developer	109%	$102,500	77
7	Development operations engineer	97%	$105,000	826
8	Product owner	76%	$90,000	589
9	Financial controller	66%	$90,000	314
10	Salesforce administrator	56%	$85,000	56

Source: Indeed

indeed

- Data related job roles are controlling comparable jobs reports released over the past a few years.



Lookup table



Computer Understanding for Lookup Table

Humans versus machines

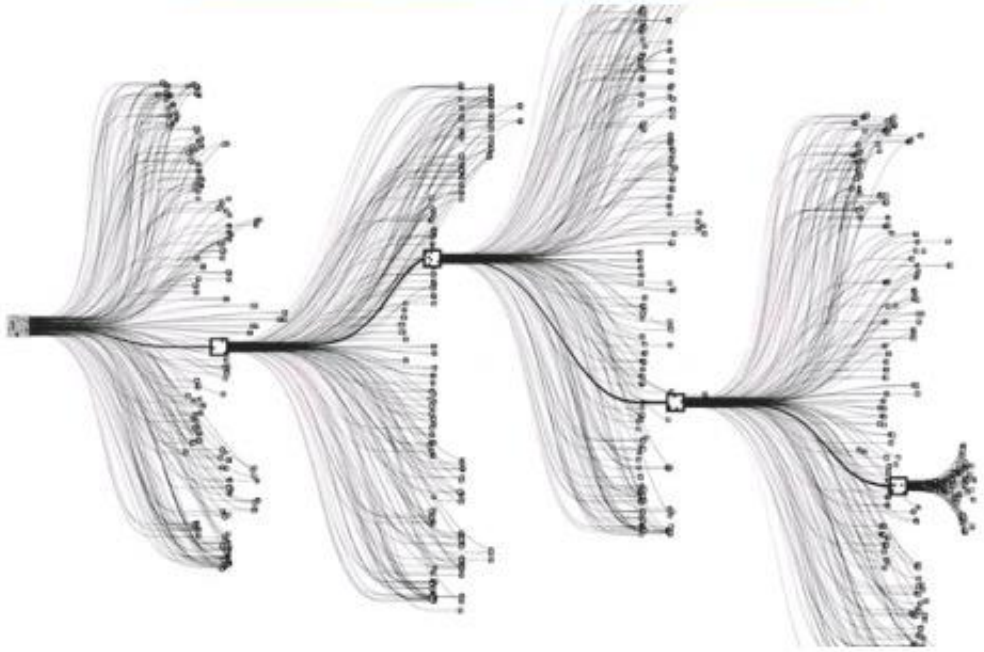


1997: Deep Blue (chess)



2011: IBM Watson (Jeopardy!)





- In 1997, Deep Blue defeated the world chess champion.
- In 2011, IBM Watson defeated two of the biggest winners at the quiz show Jeopardy
- In 2016, Google DeepMind created a program called AlphaGo, which used deep neural networks and reinforcement learning, to defeat a world champion, surprising not only the master Go player but many AI researchers as well.

Humanoid Robot and AI

- Sophia is a social humanoid robot developed by Hong Kong based company Hanson Robotics.
- Sophia was activated on April 19, 2015.
- She made her first public appearance at South by Southwest Festival in mid-March 2016 in United States.
- In October 2017, Sophia became a Saudi Arabian citizen, the first robot to receive citizenship in any country.





Any Question